SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I __U. Ananya(192424382)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

U. Ananya
Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks **/40** ***Signature of the Faculty***

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

ANALYTICAL PROBLEM SOLVING

1. Subnetting the network 192.168.20.0/24 for three labs

Given data

- Original network address: 192.168.20.0/24
- Lab A requires a /26 subnet.
- Lab B requires a /27 subnet.
- Lab C requires a /28 subnet.

Step 1: Calculate the subnet masks

A subnet mask determines which portion of an IP address identifies the network and which portion identifies the host.

Lab	**CIDR notation**	**Subnet mask**
Lab A	/26	255.255.255.192
Lab B	/27	255.255.255.224
Lab C	/28	255.255.255.240

Step 2: Calculate the number of addresses and usable hosts

The formulas are:

$$Total\ addresses = 2^{32-prefixlength}$$

$$Usablehosts = Totaladdresses - 2$$

The subtraction of two accounts for the network address and broadcast address in conventional IPv4 subnets.

Lab A — /26

$2^{32-26} = 2^6 = 64$

Usable hosts = $64 - 2 = 62$.

Lab B — /27

$2^{32-27} = 2^5 = 32$

Usable hosts = $32 - 2 = 30$.

Lab C — /28

$2^{32-28} = 2^4 = 16$

Usable hosts = $16 - 2 = 14$.

Step 3: Determine the network addresses

The block size is calculated using:

$Blocksize = 256 - lastoctetofsubnetmask$

- Lab A: $256 - 192 = 64$
- Lab B: $256 - 224 = 32$
- Lab C: $256 - 240 = 16$

Assign the subnets consecutively without overlapping.

Lab A: The first subnet starts at 192.168.20.0. With a block size of 64, its addresses run from .0 to .63.

- Network address: 192.168.20.0
- First usable host: 192.168.20.1
- Last usable host: 192.168.20.62

- Broadcast address: 192.168.20.63

Lab B: The next available address is .64. Its block size is 32, so the subnet runs from .64 to .95.

- Network address: 192.168.20.64
- First usable host: 192.168.20.65
- Last usable host: 192.168.20.94
- Broadcast address: 192.168.20.95

Lab C: The next available address is .96. Its block size is 16, so the subnet runs from .96 to .111.

- Network address: 192.168.20.96
- First usable host: 192.168.20.97
- Last usable host: 192.168.20.110
- Broadcast address: 192.168.20.111

Final result

Details	**Lab A**	**Lab B**	**Lab C**
Network address	192.168.20.0	192.168.20.64	192.168.20.96
Host range	.1 – .62	.65 – .94	.97 – .110
Broadcast address	192.168.20.63	192.168.20.95	192.168.20.111
Total addresses	64	32	16
Usable hosts	62	30	14

The remaining unused range is 192.168.20.112 – 192.168.20.255, containing 144 addresses.

Conclusion: The /24 network is divided into three non-overlapping subnets. Lab A supports 62 hosts, Lab B supports 30 hosts, and Lab C supports 14 hosts. The remaining addresses can be allocated to future requirements.

2. Identifying the subnets of two systems using /26

Given data

- System 1 IP address: 192.168.1.130
- System 2 IP address: 192.168.1.190
- Subnet mask: 255.255.255.192

Step 1: Identify the subnet mask

The subnet mask 255.255.255.192 corresponds to /26.

This means 26 bits are used for the network portion, and 6 bits are available for the host portion.

$$Hostbits = 32 - 26 = 6$$

$$Total\ addresses = 2^6 = 64$$

$$Usablehosts = 64 - 2 = 62$$

Step 2: Calculate the block size

The last octet of the subnet mask is 192.

$$Blocksize = 256 - 192 = 64$$

Therefore, the subnet ranges are:

- 192.168.1.0 – 192.168.1.63
- 192.168.1.64 – 192.168.1.127
- 192.168.1.128 – 192.168.1.191
- 192.168.1.192 – 192.168.1.255

Step 3: Identify the subnet for System 1

System 1 has the address 192.168.1.130.

The last octet, 130, falls between 128 and 191.

Therefore:

- Network address: 192.168.1.128
- Valid host range: 192.168.1.129 – 192.168.1.190
- Broadcast address: 192.168.1.191

Since 130 is within the valid host range, System 1 has a valid host address.

Step 4: Identify the subnet for System 2

System 2 has the address 192.168.1.190.

The last octet, 190, also falls between 128 and 191.

Therefore:

- Network address: 192.168.1.128
- Valid host range: 192.168.1.129 – 192.168.1.190
- Broadcast address: 192.168.1.191

Since 190 is the last usable host address, System 2 also has a valid host address.

Final result

Details	**System 1**	**System 2**
IP address	192.168.1.130	192.168.1.190
Network address	192.168.1.128	192.168.1.128

Valid host range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
Broadcast address	192.168.1.191	192.168.1.191
Usable host addresses	62	62
IP validity	Valid	Valid

Conclusion: Both systems belong to the same /26 subnet, 192.168.1.128/26. They can communicate directly within the same IP subnet, assuming there are no VLAN, firewall, or other network restrictions.

3. Subnet allocation for HR, Sales, IT, and Admin

Given data

The organization has been allocated the network block 172.16.0.0/16.

The department requirements are:

- HR: /25
- Sales: /26
- IT: /27
- Admin: /28

The /16 network contains $2^{16} = 65,536$ IPv4 addresses.

Step 1: Calculate the subnet masks and host capacities

Department	**Prefix**	**Subnet mask**	**Host bits**	**Total addresses**	**Usable hosts**

HR	/25	255.255.255.128	7	128	126
Sales	/26	255.255.255.192	6	64	62
IT	/27	255.255.255.224	5	32	30
Admin	/28	255.255.255.240	4	16	14

Step 2: Allocate the HR subnet

HR uses a /25 prefix.

$2^{32-25} = 128 addresses$

Starting at 172.16.0.0, the subnet covers the last-octet range 0–127.

- Network address: 172.16.0.0
- First usable host: 172.16.0.1
- Last usable host: 172.16.0.126
- Broadcast address: 172.16.0.127
- Usable hosts: 126

Step 3: Allocate the Sales subnet

The next available address is 172.16.0.128.

Sales uses /26, which contains 64 addresses.

- Network address: 172.16.0.128
- First usable host: 172.16.0.129
- Last usable host: 172.16.0.190
- Broadcast address: 172.16.0.191
- Usable hosts: 62

Step 4: Allocate the IT subnet

The next available address is 172.16.0.192.

IT uses /27, which contains 32 addresses.

- Network address: 172.16.0.192
- First usable host: 172.16.0.193
- Last usable host: 172.16.0.222
- Broadcast address: 172.16.0.223
- Usable hosts: 30

Step 5: Allocate the Admin subnet

The next available address is 172.16.0.224.

Admin uses /28, which contains 16 addresses.

- Network address: 172.16.0.224
- First usable host: 172.16.0.225
- Last usable host: 172.16.0.238
- Broadcast address: 172.16.0.239
- Usable hosts: 14

Final allocation table

Department	**Network address**	**Valid host range**	**Broadcast address**	**Usable hosts**
HR	172.16.0.0/25	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	172.16.0.128/26	172.16.0.129 – 172.16.0.190	172.16.0.191	62

IT	172.16.0.192/27	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	172.16.0.224/28	172.16.0.225 – 172.16.0.238	172.16.0.239	14

Step 6: Calculate the remaining unused range

Total addresses allocated:

$$128 + 64 + 32 + 16 = 240$$

Total addresses in the original network:

$$2^{32-16} = 65,536$$

Remaining addresses:

$$65,536 - 240 = 65,296$$

The last allocated address is 172.16.0.239.

Therefore, the unused range is:

172.16.0.240 – 172.16.255.255

Conclusion: The four departments have been assigned separate, non-overlapping subnets. A total of 240 addresses are allocated, and 65,296 addresses remain unused within the original /16 block. These addresses can support future expansion.

4. Subnetting 172.16.10.0/24 using a /27 subnet mask

Given data

- Original network: 172.16.10.0/24
- New subnet prefix: /27

- IP address to check: 172.16.10.78
- Second IP address to check: 172.16.10.95

Step 1: Calculate the number of subnets

The original prefix is /24, and the new prefix is /27.

Bits borrowed for subnetting:

$$27 - 24 = 3$$

Number of subnets:

$$2^3 = 8$$

Therefore, eight subnets are created.

Step 2: Calculate the number of usable hosts

The /27 prefix leaves five bits for hosts.

$$32 - 27 = 5$$

Total addresses per subnet:

$$2^5 = 32$$

Usable hosts per subnet:

$$32 - 2 = 30$$

Each subnet contains 32 addresses, of which 30 can be assigned to devices.

Step 3: Calculate the block size

The subnet mask for /27 is 255.255.255.224.

$$256 - 224 = 32$$

The eight subnets are:

Subnet number	Network address	Valid host range	Broadcast address
1	**172.16.10.0**	**.1 – .30**	**.31**
2	**172.16.10.32**	**.33 – .62**	**.63**
3	**172.16.10.64**	**.65 – .94**	**.95**
4	**172.16.10.96**	**.97 – .126**	**.127**
5	**172.16.10.128**	**.129 – .158**	**.159**
6	**172.16.10.160**	**.161 – .190**	**.191**
7	**172.16.10.192**	**.193 – .222**	**.223**
8	**172.16.10.224**	**.225 – .254**	**.255**

Step 4: Identify the subnet for 172.16.10.78

The address 78 falls within the third subnet, whose last-octet range is 64–95.

Therefore:

- Network address: 172.16.10.64
- First usable host: 172.16.10.65
- Last usable host: 172.16.10.94
- Broadcast address: 172.16.10.95

The address 172.16.10.78 is a valid host address because it lies between .65 and .94.

Step 5: Check 172.16.10.95

The address 172.16.10.95 lies within the third subnet's overall range, but it is the broadcast address.

A broadcast address is used to send packets to all hosts in that subnet. It cannot be assigned to an individual computer or device.

Conclusion: Eight subnets are created, with 30 usable hosts per subnet. The IP address 172.16.10.78 belongs to 172.16.10.64/27. The address 172.16.10.95 is in the same subnet but is reserved as the broadcast address, so it is not a valid host address.

5. IPv6 address compression, expansion, and /64 prefix

Given IPv6 address

2001:0db8:0000:0000:0000:ff00:0042:8329

IPv6 addresses are 128 bits long and are written as eight groups of hexadecimal digits separated by colons. Each group represents 16 bits.

Step 1: Apply IPv6 compression rules

IPv6 compression follows two main rules:

1. Leading zeros in each group can be removed.
2. A consecutive sequence of all-zero groups can be replaced with ::. This replacement can be used only once in an address.

Given address:

2001:0db8:0000:0000:0000:ff00:0042:8329

First, remove leading zeros from individual groups:

2001:db8:0:0:0:ff00:42:8329

Next, replace the longest consecutive sequence of zero groups with ::.

Compressed address:

2001:db8::ff00:42:8329

Step 2: Expand the compressed address

The compressed address is:

2001:db8::ff00:42:8329

There are five visible groups: 2001, db8, ff00, 42, and 8329. Therefore, :: represents three zero groups to make eight groups in total.

Expanded address:

2001:0db8:0000:0000:0000:ff00:0042:8329

This is the same as the original address.

Step 3: Determine the network prefix using /64

The /64 prefix means that the first 64 bits identify the network, while the remaining 64 bits are available for interface identifiers.

Since each IPv6 group contains 16 bits:

$64 \div 16 = 4 groups$

The first four groups are:

2001:0db8:0000:0000

Therefore, the network prefix is:

2001:db8:0:0::/64

Step 4: Calculate the number of possible host addresses

IPv6 address length:

$128 bits$

Network bits:

$64bits$

Remaining interface identifier bits:

$128 - 64 = 64bits$

Number of possible bit patterns:

$2^{64} = 18,446,744,073,709,551,616$

Thus, a /64 network contains 2^{64} possible interface identifier values.

Unlike conventional IPv4 subnetting, IPv6 does not use a broadcast address. The calculation above gives the size of the address space; practical address assignment may have additional conventions and restrictions.

Conclusion: The compressed IPv6 address is 2001:db8::ff00:42:8329, and the expanded form is 2001:0db8:0000:0000:0000:ff00:0042:8329. Its /64 network prefix is 2001:db8:0:0::/64, with 64 bits remaining for interface identifiers and 2^{64} possible values.